SQL Notes

SQL is Structured Query Language. It is a language used to get the data from database.

Data is stored in a table. Each row is a record and column is a property.

SQL database management systems are software used to execute SQL language.

- MySQL
- PostgreSQL
- SQLite
- Oracle
- DB2

Example of a Table:

Students

S. No.	Name	Last_name	Age	Marks
1	Roy	Joe	34	90
2	Lena	Dzousa	30	75
3	Ajay	Pathak	45	87
4	Ezaam	Joe	24	65
5	Devansh	Naik	48	76

SQL Commands

SELECT

1. **Select command**: It is used to fetch the data from the table.
   select name, age from Students;

WHERE

2. **Where condition:** It returns only data that match with the given condition
   select name from Students where Marks > 65;
3. **Multiple conditions:**
   AND: It returns only data when both conditions are true.
   select name from Student where Marks > 65 and Marks < 90;
   OR: It returns data only if at least one condition is true.
   select name from Student where Marks>65 or Marks<90;
4. **LIKE operator:**
   It is used for searching text patterns.
   Starts with ‘A’
   select name from Student where name like ‘A%’;
   Ends with ‘A’
   select name from Student where name like ‘%A’;
   Contains ‘A’
   select name from Student where name like ‘%A%’;
5. **Order By:**

It is used to sort the results.
Ascending Order
select name from Student order by salary;
Descending Order
select name from Student order by salary desc;

SQL generally follows the below given Execution order:

- FROM
- WHERE
- SELECT
- ORDER BY

6. **Limit:**
   It restricts the result returned.
   select name from Student limit 4;

Filtering

1. **Between**
   It is used to filter values within a range.
   select name from Student where marks between 75 and 90;
   This includes value including 30 and 45
   It works similar to marks >= 75 and marks<=90;
2. **IN**
   It is used to fetch values that includes multiple exact values.
   select name from Student where last_name in ('Joe' , 'Pathak');
   This works similar to where name = 'Joe' or name = 'Pathak';
3. **Not like**
   It is used like a negative condition.
   select name from student where last_name not like 'J%';

   select name from Student
   where marks
   between 65 and 89
   and last name not like 'J%'
   order by name desc
   limit 2;

Aggregate Functions

It summarizes the data.

1. **Count()**
   This counts the rows.
   select count(*) from Student;
2. **Average()**
   This calculates the average.
   select avg(marks) from Student;
3. **Sum()**
   This calculates the sum.

select sum(marks) from Students;

4. **Max()**

   This gives the maximum value.

   select max(marks) from Students;

5. **Min()**

   This gives the minimum value.

   select min(marks) from Students;

GROUP BY

- It groups rows with same value.
- Aggregate functions work on each group.
- Without Group By, SQL cannot calculate the summaries of a group.

Orders

Item_Id	Items	Amount
1	Printer	2000
2	Keyboard	5000
3	Mouse	600
4	Mousepad	300
5	Keyboard	2000
6	Mouse	790

select Items, sum(amount) from Orders group by Items;

select *
from table_name
where condition
group by column
order by column desc
limit 5;